Behaviour change interventions to increase physical activity in hospitalized patients: A systematic review, meta-analysis and meta-regression

Supplementary Materials

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Appendix 10. Meta-analysis: Length of stay in rehabilitation settings. Note. GRADE certainty of evidence moderate downgraded due to wide confidence intervals

Appendix 11. References cited in Appendices

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract p1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	pp3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	p4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	PROSPERO (CRD42020170445 p4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5, Appendix 3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Appendix 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	p5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	p6

Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	p6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	p6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	pp6-7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6-7

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	p7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	p7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	pp7-8, Figure1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	p9, Appendix 6
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Figures 2-4, Appendices 9-10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	pp9-11, figures 2-4, Appendices 9-10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	pp11-12, plus figure

			captions
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11, Appendices 7-8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	pp11-12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	p13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	p14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Title page

Appendix 1. PRISMA checklist

Appendix 2. Literature search strategy MEDLINE

1. ward.mp (58905)

2. Hospitalization/ or hospital*.mp (303514)

3. Inpatients/or inpatient.mp. (113919)

4. in-hopsital.mp (85521)

5. “admitted to hospital”.mp. (11204)

6. “hospital stay”.mp. (74010)

7. postsurgical.mp. (13833)

8. post-surgical.mp. (8004)

9. Postoperative Period/ or Postoperative care/ or postoperative.mp. (780623)

10. post-operative.mp. (59232)

11. Hospitals, Rehabilitation/ or Rehabilitation Centers/ or Rehabilitation/ (25545)

12. Accelerometry/ or accelerometr*.mp. (17266)

13. pedometer*.mp. or Actiography/ (5783)

14. “steps per day”.mp. (918)

15. “daily steps”.mp. or Monitoring Ambulatory/ (8525)

16. “step count”.mp. (976)

17. wearable technology.mp. or Wearable Electronic Devices/ (2476)

18. activity monitor*.mp. (3698)

19. “activity count”.mp. (116)

20. METs.mp. (10654)

21. metabolic equivalent *.mp. (2991)

22. inertial measurement units.mp. (

23. ambulatory monitoring.mp. or Monitoring Ambulatory/

24. (activity adj3 track*).mp.[mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique idenitifer, synonyms] (1562)

25. Fitness Trackers/ or fitness tracker*.mp. (612)

26. fitbit*.mp. (561)

27. actigraph.mp. (2640)

28. activpal*.mp. (445)

29. positional activity logger.mp. (2)

30. Randomized Controlled Trial/ or Clinical Trial/ or trial.mp. (1174028)

31. cluster*.mp. (232953)

32. random*.mp. (1347890)

33. activity count*.mp. (1017)

34. monitoring, ambulatory.mp. or Monitoring, Ambulatory/ (17688)

35. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 (1329473)

36, 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 33 or 34 (58870)

37. 30 or 31 or 32 (2066448)

38. 35 and 36 and 37 (564)

Appendix 3. Eligibility criteria

Inclusion	Exclusion
P: People of any age hospitalized for any physical or mental health condition, including: • acute hospital care • inpatient rehabilitation • inpatient mental health care.	People receiving outpatient, ambulatory care or community services and those attending emergency departments. People living in residential care
I: Behavioral interventions directed at increasing levels of physical activity. Behavioral interventions include those described in the CALO_RE taxonomy.[1]	Non-behavioral interventions or those not described in the CALO_RE taxonomy; surgical interventions; pharmacological interventions.
C: Hospitalized patients who did not receive the health behavior intervention.	
O: Physical activity using objective measures (e.g. accelerometer, pedometer, fitness tracker). Eligible measures include daily energy expenditure (e.g. daily steps, metabolic equivalent of tasks (METS), activity counts, daily minutes of at least moderate-intensity physical activity) during hospital admission. Note: Measures of care quality including other patient outcomes, service outcomes (e.g. length of stay), safety and patient experience to be extracted, but data on primary outcome must be present to meet inclusion criteria.	Measures of activity limitation or participation restriction Measures of physical activity obtained by self-report
R: Randomized controlled trials of any sort including: • Pilot and feasibility randomized controlled trials • Cluster randomized controlled trials • Stepped Wedge Cluster randomized controlled trials	Non-randomized study designs
Peer reviewed journal articles written in English	Theses, abstracts Publications in other languages

Appendix 4. Description of experimental group interventions according to TIDieR (template for intervention description and replication)

Lead author	Why: Behavior change techniques[a]	What materials	What procedures	Who provided	How	Where (setting)	When/how much (dose)	Tailoring	Modif-ications	Fidelity checking measures	Actual fidelity
Atkins[2]	Goals (behav); Self-monitor	Pedometer	Exercise program Steps recorded	Physio + Family/ Nursing staff	Face-to-face Individual or group	Rehab ward	Daily feedback on steps and exercises	Individual exercises, Standard feedback	Not reported	Not reported	Not reported
Au[3]	Goals (behav); Goals review (behav)	Exercise band, mat, activity ball, manual, video	Exercise program + supporting behavior change strategies	Not reported	Individual Face-to-face then phone/ email	Acute ward (surgery)	4-8 weeks prior to surgery, 60 minutes, 3-4 days/week.	Individual exercises.	Not reported	Adherence: weekly phone/email contact	Adherence 84% (aerobic) 79% (resistance)
Dall[4]	Info (general); Feedback	Tablet for feedback Accelerometer Written information	Mobility assessed. Accelerometer feedback on bedside screens	Physio	Face-to-face Individual	Acute ward (medical)	1 x information session. Feedback every 5 minutes	Individual program based on walking score	Not reported	Not reported	Not reported
Despond[5]	Enviro restruct	Day clothes	Preadmission instruction to wear day clothes in hospital	Researcher	Face-to-face Individual	Instructions at home, Acute ward (surgical)	Wore clothes during waking hours	Standard	Not reported	Days in day clothes counted	20/28 patients in day clothes day 2 post surgery
Dorsch[6]	Feedback	Accelerometer	Verbal and visual accelerometer feedback	Therapists	Face-to-face Individual	Rehab ward	3 x weekly feedback	Individual feedback	Not reported	Not reported	Not reported

Fiore[7]	Action planning; Instruct (Where/ when); Instruct (How); Practice	Mobilization Goal booklet	Instructions pre-op; post op assisted walking and goal reinforcement	Nurse pre-op; Physio or student on ward	Face-to-face Individual	Pre-op clinic Acute ward (surgical)	1 pre-op clinic appointment ; Prompts/ assistance 3 x day on ward	Individual walking target	Not reported	Self-reported time out of bed; time with assisted mobilization	Median time receiving assisted mobilization 116 min
Jonsson[8]	Info (Indiv); Goals (behav)	Not reported	Pre-op information; Post-op encouraged to walk	Physio + usual care nursing	Face-to-face Individual	Acute ward (surgical)	Daily physio 20-30 min for first 2 days. 6 days/week	Individual physio	Not reported	Not reported	4 patients discharged before post-op day 3
Kanai[9]	Goals (behave); Goals (outcomes); Goals review (outcomes); Rewards; Self-monitoring; Feedback	Accelerometer Exercise calendar	Daily physical activity goals and Fitbit feedback	Physio / Occupational therapist	Face-to-face Individual	Acute ward (medical)	Supervised rehab 5- 6 x week, 40-120 min sessions. Time on activity goals not reported	Individual rehab program and activity feedback	Not reported	Not reported	1 patient did not wear accelerometer
Lieberman[10]	Goals (outcome) Practice	Bedside signs	500 step goal set, staff reminders and ward signage	Doctor, Nurse	Face-to-face Individual	Acute ward (surgical)	Staff reminders to walk at every encounter	Standard	Not reported	Not reported	Not reported
Mansfield[11]	Info (Others); Goals (behav) Goals review (behav);	Accelerometer	Feedback to physios of daily walking; review of walking goals	Physio	Face-to-face Individual	Rehab ward	1-hour daily physio; daily step count feedback on & goal review	Individual. Goals adjusted from activity reports	Not reported	Not reported	7 patients < 3 days of activity data

	Feedback										
Moreno[12]	Info (indiv); Instruction (how) Cues	Booklet	Booklet plus discussion with physio	Physio	Face-to-face Individual	Acute ward (medical)	20 min Face-to-face physio + usual care	Standard	Not reported	Not reported	Potential contamination (shared ward)
Papaspyros[13]	Enviro restruct	Bedside tv, phone, gaming & internet	Access to bedside entertainment	Not reported	Individual	Acute ward (surgical)	Screen covered all day from day 2	Standard	Not reported	Not reported	Not reported
Peel[14]	Goals (behav); goals review (behav); Feedback	Accelerometer	Feedback on walking, and subsequent mobility goals	Physio	Face-to-face Individual	Rehab ward	Daily for 28 days	Individual mobility goals	Device changed (ALIVE to ActivPal)	Staff trained in use of accelerometer data	Not reported
Swank[15]	Action planning; Problem solving Graded tasks Instruct (how)	Activity program	Usual care rehab + Patient Directed Activities Program	Therapist	Face-to-face Individual	Rehab ward	3 hrs usual therapy + 1 hr directed activities daily	Individual activities, moderate intensity	Not reported	Activity sessions recorded by therapists	Mean 23 sessions of 28 days in rehab
Taraldsen[16]	Goals (outcome)	Individual rehab plan.	Not reported	Physio and Nurse	Face-to-face Individual	Acute ward (surgical)	Regular meetings from day 1	Individual plan and goals	Completed according to protocol	Not reported	Not reported
Teodoro[17]	Info (general); Info (indiv); Info (others); Goals (behav); Graded tasks; Modelling; Practice	Video, white board, walking reminder card	Brief video about importance of walking and mobility instructions.	Researcher	Face-to-face Individual	Acute ward (medical-surgical)	Watched video Daily goals and reminders for 3 days	Individual	Not reported	Training using pedometer	Not reported

Van der Walt[18]	Goals (behav); Feedback	Accelerometer	Physical activity goal & visual feedback	Researcher	Face-to-face Individual	Acute ward (surgical)	Daily step goal on ward, constant feedback	Individual	Not reported	Not reported	Not reported
Wolk[19]	Goals (behav); Feedback	Accelerometer	Feedback provided on step count target	Doctor or study Nurse	Face-to-face Individual	Acute ward (surgical)	2x daily activity feedback 5 days post-op	Standard step count targets	Not reported	Monitors checked regularly	Not reported

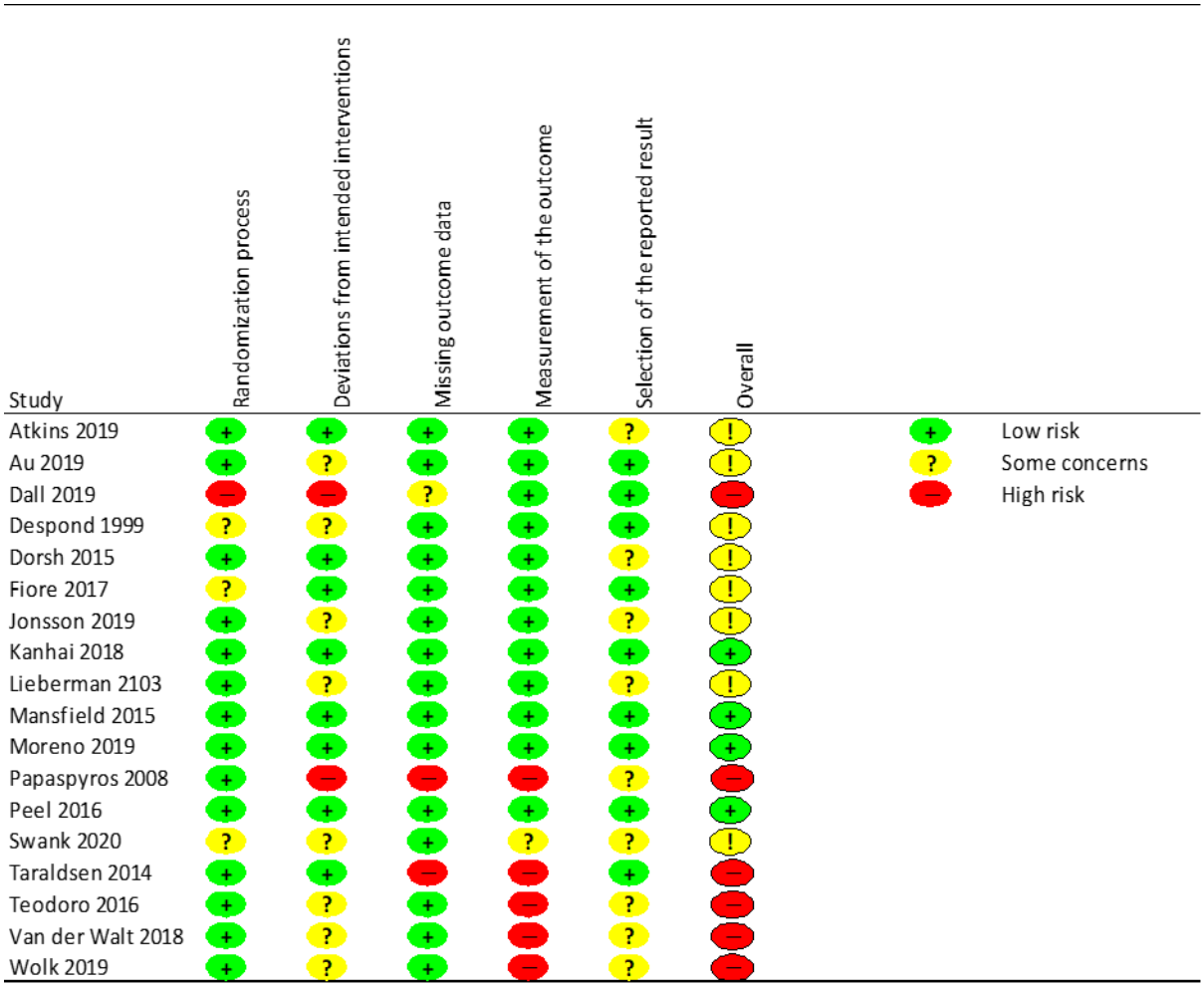
[a] Behavioor change techniques according to CALO-RE taxonomy.[1] *Info (general)* - Provide information on consequences of behaviour in general; *Info (indiv)* - Provide information on consequences of behaviour to the individual; *Info (others)* - Provide normative information about others' behavior; *Goals (behav)* - Setting behavioral goals; *Goals (outcomes)* -Setting outcome goals; *Action planning* - Planning what will be done; *Problem solving* - Planning for overcoming barriers; *Graded tasks* - Breaking down the behavior into smaller tasks; *Goals review (behav)* - Prompting review of behavioral goals; *Goals review (outcome)* -Prompting review of outcome goals; *Rewards* - reinforcement of target behavior; *Self-monitor* - Prompting of self-monitoring of behavioral outcome; *Feedback* - Providing feedback on performance; *Instruct (Where & when)* - Instructions on where and when to perform the behavior; *Instruct (How)*- Instructions on how to perform the behavior; *Modelling* - Modelling or demonstration of the behavior; *Cues* - Teach to use prompts/ cues; *Enviro Restruct* - restructuring the environment to be more supportive of the behavior; *Practice* - prompting the person to repeat/rehearse the behavior.

Appendix 5. Description of comparison group interventions according to TIDieR

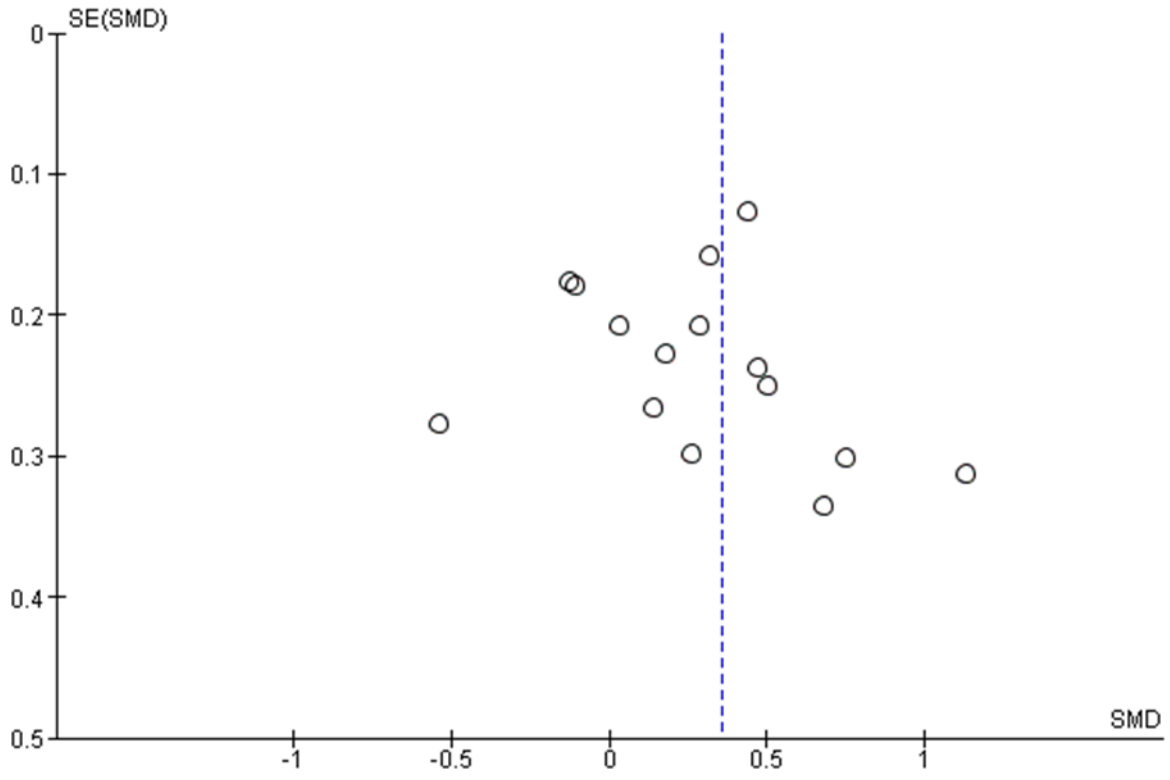
Lead author	What provided	Who provided	How	Where (setting)	When/how much (dose)	Tailoring	Modifica-tions	Fidelity checking measures	Actual fidelity
Atkins[2]	Usual care (exercise and walking); pedometer with step count hidden	Physio +Family/nursing staff	Face-to- face Individual or group	Rehab ward	Daily exercise with verbal encourage-ment	Individualized exercise plan	Not reported	Not reported	7 patients aware of their step count
Au[3]	Usual care (exercises + written advice)	Urology nurse	Individual	Acute ward (surgery)	Pelvic floor exercises 60-180 reps per day	Standardized pelvic floor program	Not reported	Adherence: Weekly phone/email contact	Not reported
Dall[4]	Information; Accelerometer with no feedback	Physio	Face-to-face Individual	Acute ward (medical)	Single information session	Not reported	Not reported	Not reported	Not reported
Despond[5]	No preadmission clothing instructions given for when in hospital	Researcher	Face-to-face Individual	Instructions at home Acute ward (surgical)	Monitored continuously	Standardized	Not reported	Days in plain clothes counted	26/29 wore hospital gown until day of discharge
Dorsch[6]	Verbal feedback on walking	Therapist	Face-to-face Individual	Rehab ward	3x weekly feedback	Individual feedback	Not reported	Not reported	Not reported
Fiore[7]	Usual care (Instructions pre-op; assistance post-op to get out of bed on first occasion only)	Nurse	Face-to-face Individual	Pre-op clinic Acute ward (surgical)	N/a	Nurse could assist if ward team deemed necessary or patient requested	Not reported	Self-reported time out of bed; time with assisted mobilization	Achievement of mobilization target 36% on day of surgery; 47% on day 3 post op

Jonsson[8]	Usual care (walking, pain management and nursing care)	Usual care nursing	Not reported	Acute ward (surgical)	Not reported	Not reported	Not reported	Not reported	6 patients discharged before post-op day 3
Kanai[9]	Rehab (No accelerometer)	Physio / Occupational therapist	Face-to-face Individual	Acute ward (medical)	Supervised rehab 5-6 x week, 40-120 min sessions.	Individual rehab program	Not reported	Not reported	Not reported
Lieberman[10]	Usual care (no extra encouragement).	Doctor, Nurse	Not reported	Acute ward (surgical)	Not reported	Not reported	Not reported	Not reported	Not reported
Mansfield[11]	Rehab and activity monitor (no activity feedback)	Physio	Face to face Individual	Rehab ward	1-hour daily physio	Individual goals 'as needed'	Not reported	Not reported	3 participants < 3 days of activity data
Moreno[12]	Usual clinical care (no written or verbal activity advice)	Not reported	Not reported	Acute ward (medical)	Not reported	Not reported	Not reported	Not reported	Potential contamination shared ward
Papaspyros[13]	Access to communal ward entertainment	None	Individual	Acute ward (surgical)	Patients observed post-op day 2 until discharge	Not reported	Not reported	Not reported	Not reported
Peel[14]	Usual care and rehab (no data on walking times)	Physio	Face-to- face Individual	Rehab ward	4 weeks from study entry	Individual mobility goals	Device changed (ALIVE to ActivPal)	Staff trained in use of accelerometer data	Not reported
Swank[15]	Usual care rehab	Therapists	Face-to-face Individual	Rehab ward	3 hrs usual therapy	Not reported	Not reported	Not reported	Not reported

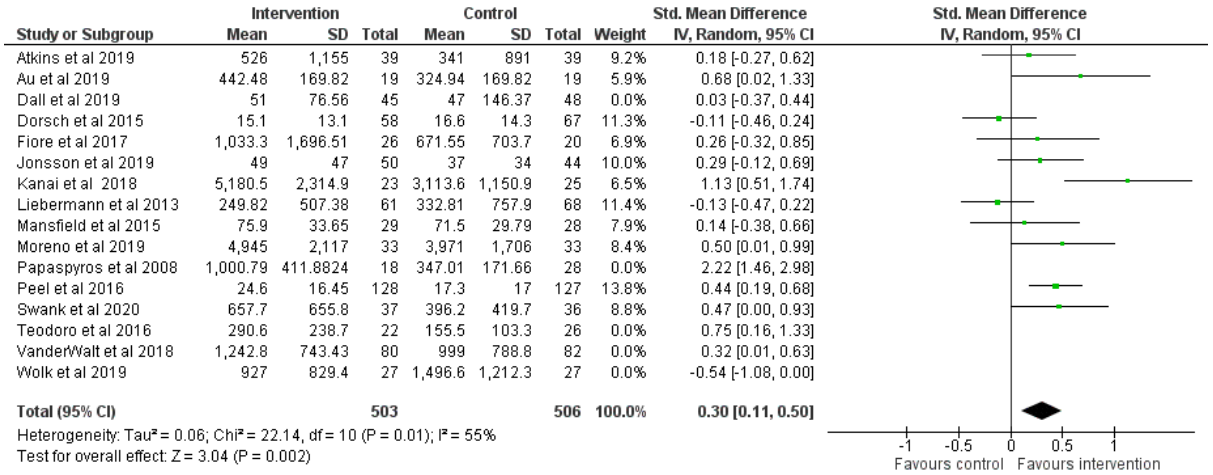
Taraldsen[16]	Usual orthopedic care	Physio + Clinical team	Face-to- face Individual	Acute ward (surgical)	Not reported	Not reported	Completed according to protocol	Not reported	Not reported
Teodoro[17]	Usual care	Nurse	Not reported	Acute ward (medical-surgical)	Not reported	Not reported	Not reported	Not reported	Not reported
Van der Walt[18]	Accelerometer (No feedback)	Researcher	Face-to- face Individual	Acute ward (surgical)	Daily steps counted ward	Not reported	Not reported	Not reported	Not reported
Wolk[19]	Activity monitor (no feedback)	Study nurse	Face-to-face Individual	Acute ward (surgical)	5 days post-op	Not reported	Not reported	Monitors checked regularly	Not reported



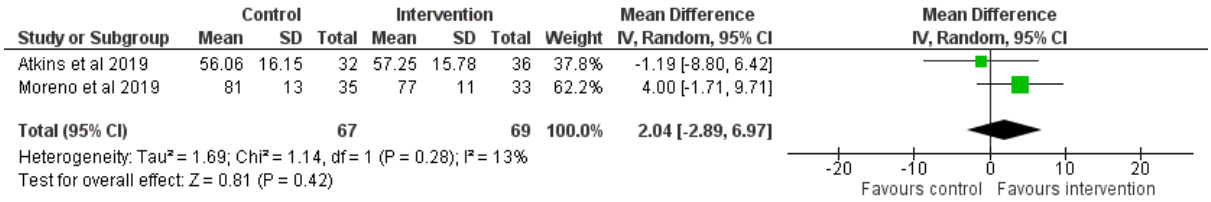
Appendix 6. Risk of bias



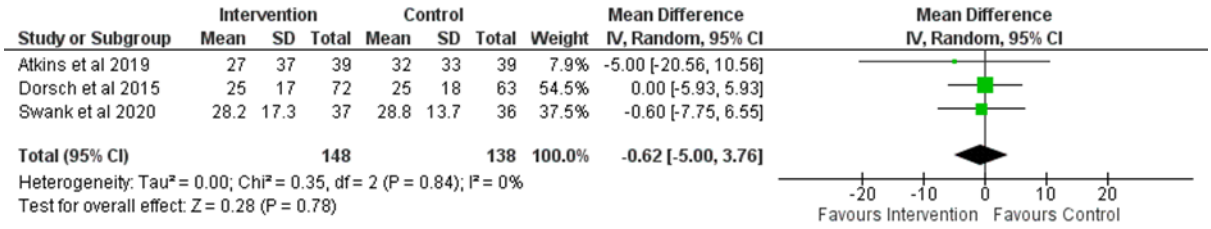
Appendix 7. Physical activity meta-analysis funnel plot



Appendix 8. Meta-analysis: Physical activity - sensitivity analysis excluding trials with a high risk of bias



Appendix 9. Meta-analysis: De Morton Mobility Index (DEMMI). Note. GRADE certainty of evidence rated as high



Appendix 10. Meta-analysis: Length of stay in rehabilitation settings. Note. GRADE certainty of evidence moderate downgraded due to wide confidence intervals

Appendix 11. References

1. Michie S, Ashford S, Sniehotta FF, et al. A refined taxonomy of behaviour change techniques to help people change their physical activity and healthy eating behaviours: the CALO-RE taxonomy. *Psychol Health.* 2011;26(11):1479-1498. doi: 10.1080/08870446.2010.540664

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